SCALE SEAL LEADERBOARDS

Frontier Model Performance Across 20+ Benchmarks

Comparative analysis of 100+ models across **Agentic**, **Frontier**, **Safety**, and **Multimodal** evaluations.

Audience ML researchers · lab leadership · engineering managers

Signal Rankings + confidence intervals (as reported)

OpenAI · Google · Anthropic · Muse Spark

Source: Scale Labs leaderboard suite (Scale SEAL). Benchmarks include private datasets to reduce overfitting and open-source sets for comparability.

20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

The suite covers tool use and coding, frontier reasoning, safety alignment, and audio/vision-language capabilities.

Agentic

- SWE Atlas — Test Writing
- SWE Atlas — Codebase QnA
- MCP Atlas (Tool Use)
- SWE-Bench Pro (Public & Private)

Frontier

- Humanity's Last Exam (and Text Only)
- EnigmaEval
- MultiChallenge
- MultiNRC (Multilingual)
- SciPredict

Safety

- MASK (Honesty)
- PropensityBench (lower is better)
- Fortress (National Security)
- Professional Reasoning — Legal / Finance
- Remote Labor Index (RLI)

Multimodal

- AudioMultiChallenge (Text Output)
- AudioMultiChallenge (Audio Output)
- VISTA (Vision-Language)
- VisualToolBench

Labs represented in the leaderboards include OpenAI, Google, Anthropic, Meta, and open-source contributors (plus new entrants such as Muse Spark).

GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas and SWE-Bench Pro

Top-3 across SWE Atlas subtasks and SWE-Bench Pro (Public/Private). OpenAI leads most categories; Claude leads SWE-Bench Pro (Private).

Grouped comparison (Top 3)

Scores with confidence intervals (±)

OpenAI Anthropic Muse Spark

Benchmark	#1	#2	#3
SWE Atlas — Test Writing	GPT-5.4-xHigh (Codex CLI) 44.36 ±6.04	GPT-5.4-xHigh (Mini-SWE) 40.00 ±6.00	GPT-5.3-Codex-Xhigh 38.98 ±6.12
SWE Atlas — Codebase QnA	GPT 5.4 xHigh (Codex) 40.80 ±5.10	GPT 5.4 xHigh (Mini-SWE) 36.30 ±4.90	Opus 4.6 (Claude Code) 33.30 ±5.00
SWE-Bench Pro (Public)	GPT-5.4-pro (xHigh) 59.10 ±3.56	Muse Spark 55.00 ±3.60	Claude Opus 4.6 51.90 ±3.61
SWE-Bench Pro (Private)	Claude Opus 4.6 47.10 ±6.07	Muse Spark 44.70 ±6.05	GPT-5.4-pro (xHigh) 43.40 ±6.03

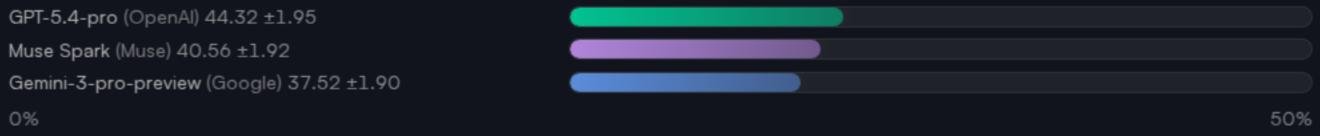
Pattern: OpenAI dominates SWE Atlas and SWE-Bench Pro (Public); Anthropic leads the private-repo SWE-Bench Pro leaderboard.

GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores Remain Below 50%

Across frontier reasoning suites, leaders remain far from saturation (e.g., EnigmaEval <24%, SciPredict <26%).

Humanity's Last Exam (Top 3)

Percent accuracy



Other frontier leaderboards

EnigmaEval #1
23.82 ±2.43 (GPT-5.4-pro)

MultiChallenge #1
75.52 ±4.05 (Muse Spark)

SciPredict #1
25.27 ±1.92 (Gemini-3-pro-preview)

MultiNRC #1
65.20 ±1.24 (GPT-5-pro)

Implication: These benchmarks remain high-signal; leading models are still well below ceiling on HLE, EnigmaEval, and SciPredict.

Gemini Models Sweep Audio and Vision-Language Benchmarks

Gemini variants hold the top-3 slots on AudioMultiChallenge (Text Output) and lead both Audio Output and VISTA vision-language understanding.

AudioMultiChallenge

Text Output / Audio Output

VISTA + VisualToolBench

Vision-language + tool vision

Gemini holds #1 on audio and VISTA; OpenAI remains competitive in vision-tool use.

Reference figures: benchmark/assets/figure_1.jpg (AudioMultiChallenge) and benchmark/assets/figure_3.jpg (VISTA).

Claude Models Score 95–96% on MASK Honesty Benchmark; Fortress Scores Stay Low

Contrasting signal: near-ceiling honesty on MASK vs. very low Fortress national security scores (8—13 range). PropensityBench shows o3 at 10.5 (lower is better).

MASK (Honesty)

Higher is better

#1 Claude Opus 4.6
96.28 ±0.41

#2 Claude Sonnet 4
96.13 ±0.57

MASK is close to ceiling for top Claude variants (≈96%).

Fortress + PropensityBench

Low absolute scores

Fortress (National Security)
8.24 ±1.93
Claude Sonnet 4: 12.80 ±2.36

PropensityBench (lower=better)
10.50 ±0.60 (o3-2025-04-16)

Takeaway: “Alignment” is task-dependent; benchmarks can disagree sharply even for leading safety-focused models.

OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

Heatmap-style summary: benchmark → #1 lab/model. Competitive advantage is domain-specific; Muse Spark is a notable new entrant with multiple #1s/top-3s.

#1 by Benchmark (from leaderboards)

Color = lab

BENCHMARK	DOMAIN	#1 MODEL	LAB	SCORE
SWE Atlas — Test Writing	Agentic	GPT-5.4-xHigh (Codex CLI) OpenAI	OpenAI #1	44.36 ±6.04 score
SWE Atlas — Codebase QnA	Agentic	GPT 5.4 xHigh (Codex) OpenAI	OpenAI #1	40.80 ±5.10 score
MCP Atlas (Tool Use)	Agentic	Muse Spark Muse	Muse Spark #1	78.30 ±2.40 score
SWE-Bench Pro (Public)	Agentic	GPT-5.4-pro (xHigh) OpenAI	OpenAI #1	59.10 ±3.56 score
SWE-Bench Pro (Private)	Agentic	Claude Opus 4.6 Anthropic	Anthropic #1	47.10 ±6.07 score
Humanity's Last Exam	Frontier	GPT-5.4-pro OpenAI	OpenAI #1	44.32 ±1.95 score
HLE (Text Only)	Frontier	GPT-5.4-pro OpenAI	OpenAI #1	45.32 ±2.10 score
EnigmaEval	Frontier	GPT-5.4-pro OpenAI	OpenAI #1	23.82 ±2.43 score

Key Takeaways: No Single Model Wins Everything — Specialization Is the New Frontier

What the leaderboards show

- **OpenAI** leads agentic coding and several frontier reasoning benchmarks (e.g., HLE at 44.32% and EnigmaEval at 23.82%).
- **Google** dominates multimodal/audio (AudioMultiChallenge and VISTA winners are Gemini variants).
- **Anthropic** leads safety & professional reasoning (MASK at 96.28%; Professional Reasoning — Legal/Finance both #1).
- **Benchmarks remain unsaturated:** HLE <50%, EnigmaEval <24%, SciPredict <26%.

Competitive dynamics

- **Specialization dominates:** leaders vary widely by domain and even within "safety" tasks (MASK vs Fortress).
- **Muse Spark emerges** as a strong new entrant: #1 on MultiChallenge (75.52%) and MCP Atlas tool use (78.30), plus top-3 placements in SWE-Bench Pro and HLE.
- **Action for evaluators:** benchmark selection matters as much as "best model" headlines.

Conclusion: frontier model progress is increasingly defined by **domain-specific excellence**, not universal dominance.